ABSTRACT

Flood is a significant environmental problem in Sonitpur Assam due to its proximity to major rivers like Brahmaputra and its tributary. This report majorly focuses on analyzing this flood vulnerability using GIS techniques. The study integrates digital elevation model historical flood record and census data to determine high risk zones and thus, highlighting critical areas which are vulnerable to flooding. This report also tries to highlight mitigation strategies such as improved in drainage systems, floodplain zoning and early warning systems which could help in mitigating floods. The study underscores the necessity of active flood management to reduce disaster impact and thus enhancing resilience.

INTRODUCTION

Quantum Geographic Information System (QGIS) is a specialized information system designed for the input, storage, retrieval, manipulation, analysis, and output of geographically referenced or geospatial data. It supports decision-making processes related to various land related planning, natural resource management, environmental conservation, transportation, urban infrastructure, and other administrative functions. QGIS is a software that has become a vital tool in disaster management in an era where natural disaster frequency and intensity are rising and decision-makers can gain important insights from QGIS, a technology that records, evaluates and displays spatial and geographic data when planning for, responding to and recovering from disasters.

While talking about natural disasters in India, flood is one of the most prominent disasters which broadly affects various terrains throughout the country. A very prominent example can be floods in Assam. Assam, with its extensive river network, is a perennial victim of natural disasters like floods and soil erosion. The Brahmaputra and Barak Rivers, fed by over 50 tributaries, are the primary causes of these recurring calamities(***District Disaster Management Authority Sonitpur | Sonitpur | Government of Assam, India***, n.d.). According to the Government of Assam, almost 40% of Assam's land is flood-prone, nearly four times the national average (***Flood & Erosion Problems | Water Resources | Government of Assam, India***, n.d.). Historical data shows a troubling trend of rising flood impacts. From the 1950s to the early 2000s, the average annual flooded area and affected cropland increased. The number of affected people surged from 860,000 to over 4.5 million in the early 2000s. (***India: What's Really Behind Assam's Worsening Floods?***, 2024b)

Although Assam experienced many devastating floods in its history, the catastrophic floods became frequent phenomena after the 1950s (earthquake) as disastrous floods occurred in 1954, 1962, 1972, 1977, 1984, 1988, 1998, 2002, 2004, 2012 and recently, in 2019 and 2020(***A Historical Understanding of Assam's Floods***, 2022). These repeated devastation in the districts of Assam underscores the need for long-term flood mitigation strategies. Dr. Partha J. Das said, "Global warming intensifies the monsoon and increases extreme rainfall events in Assam, amplifying flood

impacts with local environmental degradation"(***India: What's Really Behind Assam's Worsening Floods?***, 2024b).

The utilization of spatial representation possible due to QGIS enables emergency responders to identify vulnerable people and allocate resources strategically, facilitating risk assessment by anticipating and making plans for possible calamities by pinpointing locations vulnerable to certain risks by analyzing historical data, It can also help in emergency response planning by evacuation routes, emergency shelters and supply distribution hubs. Also helps in coordination and effective communication which are critical elements of disaster management success and lastly in post-disaster recovery efforts by comparing pre- and post-disaster data, identifying hotspots I.e most severely affected areas, prioritizing response there, using maps to optimize relief distribution and mapping safe zones and critical infrastructure.

This assignment is purely focused on the Sonitpur district of Assam. We will use QGIS to access flood vulnerability in its major watershed regions and develop insights from it using visualisations.

STUDY AREA

Sonitpur, in Assam has an area of around 5200 sq km according to the 2011 census . Brahamputra and its tributaries make the major rivers in this area. Being under a sub-tropical zone, the area enjoys a hot and humid climate with an average temperature of 29 degrees celsius. Rain in this area is both a boon and a bane. Boon because it provides natural irrigation to the fields while bane because many-a-times it causes rivers to overflow and flood nearby areas. The administrative headquarter of Sonitpur is in Tezpur. There are 5 revenue circles or tehsils of Sonitpur namely, Tezpur namely Tezpur, Dekhiajuli, Thelamara, Chariduar and Naduar. According to the 2011 census the total population of Tezpur is 19,24,110 with a 90.96% population residing in rural areas with a sex ratio of 104.65.

METHODOLOGY

This analysis aimed at three major goals:

- Watershed Delineation
- Stream Network Analysis
- Flood Vulnerability Assessment

The dataset required were as foll:

- India ADM Shapefile (country to village levels)
- Area DEM file

- Flood vulnerability status : Flood contingency report of the district by Government of Assam
- Google maps OSM (Open Street Map)
- Census Data

Steps:

Watershed Delineation

- Clipping Sonitpur district using ADM files and clipping method of geoprocessing tools
- Loading DEM file using SRTM downloader.
- Used Fill Sinks(Wang & Liu) tool for correcting depressions.
- Strahler DEM file generation and order fixing using Strahler in processing toolbox in SAGA
- Adjusting formatting through Symbology to make it visible by making changes like Singleband to Single Band pseudocolor for effective visualisations.
- Taking coordinates using 'Copy coordinate' for desired watershed area
- Using Upslope tool in SAGA under processing toolbox and paste copied coordinates to delineate watershed area
- Using Symbology its band colour have been changed for effective visualisations
- Using point feature and toggle editing to specify sub-watersheds.
- Adding x and y coordinates to the attribute table using $x and $y function.
- Running Upslope again as a batch process and copying the coordinates to delineate subwatersheds
- Changed symbology for better visualisations.
- Using Polygonise tool for changing this raster to vector file.
- Then merging the generated files using Merge Vector Layers tool.
- Adding area to the attribute table of merged using $area/1000000 to get area in sq.km. by keeping at decimal places with precision of three.
- This process is repeated for another watershed

Output:

- Delineated two major watersheds of the selected district using a Digital Elevation Model (DEM).
- Identified and extracted sub-watersheds within the main watershed.
- Calculated the total area of the main watershed and sub-watersheds in square kilometers.

Stream Network Analysis

- After filling sinks in above step, Click on Channels and Network analysis in the SAGA tool in processing toolbox
- Click on Channels and its respective box.
- Select the respective order for proper visualisation. Use the same as the order of Strahler visualisation. In this case the band was from 1-11 and a band of 8 or above is used for visualisation.
- Save it as a "Channels" file.
- Clip the watershed final shp file with the channels using the geoprocessing tool box.
- Stream Network is ready.
- Go to labels in the Channels section. Select "Order" in the drop down menu. The streams will be classified under Stahler order.
- The attribute table would contain individual lengths of stream.
- Go to vector>>Analysis Tools>>Basic statistics for fields>>Select "Length"(which was calculated)>>Run
- The output html file has a total length of streams.

Output:

- Extracted the stream network within the watershed
- Classified the streams based on Strahler's stream order.
- Determined the total length of all streams in the main watershed and its sub-watersheds.

Flood Vulnerability Assessment

- Clipped the watershed area with ADM 5 of district map of India using geoprocessing tool
- Attribute table lists down the total number of villages in the area
- Flood prone zones identified based on Flood contingency report of the Sonitpur district by Government of Assam
- Area calculated using adding a column of area using field calculator in the clipped file. Then using $area/1000000 we calculated area in sq.km
- For total area, Go to vector>>Analysis Tools>>Basic statistics for fields>>Select "Area"(which was calculated)>>Run
- The output html file has a total area under vulnerability.
- Identified areas are highlighted based on high-risk zones and vulnerable areas.

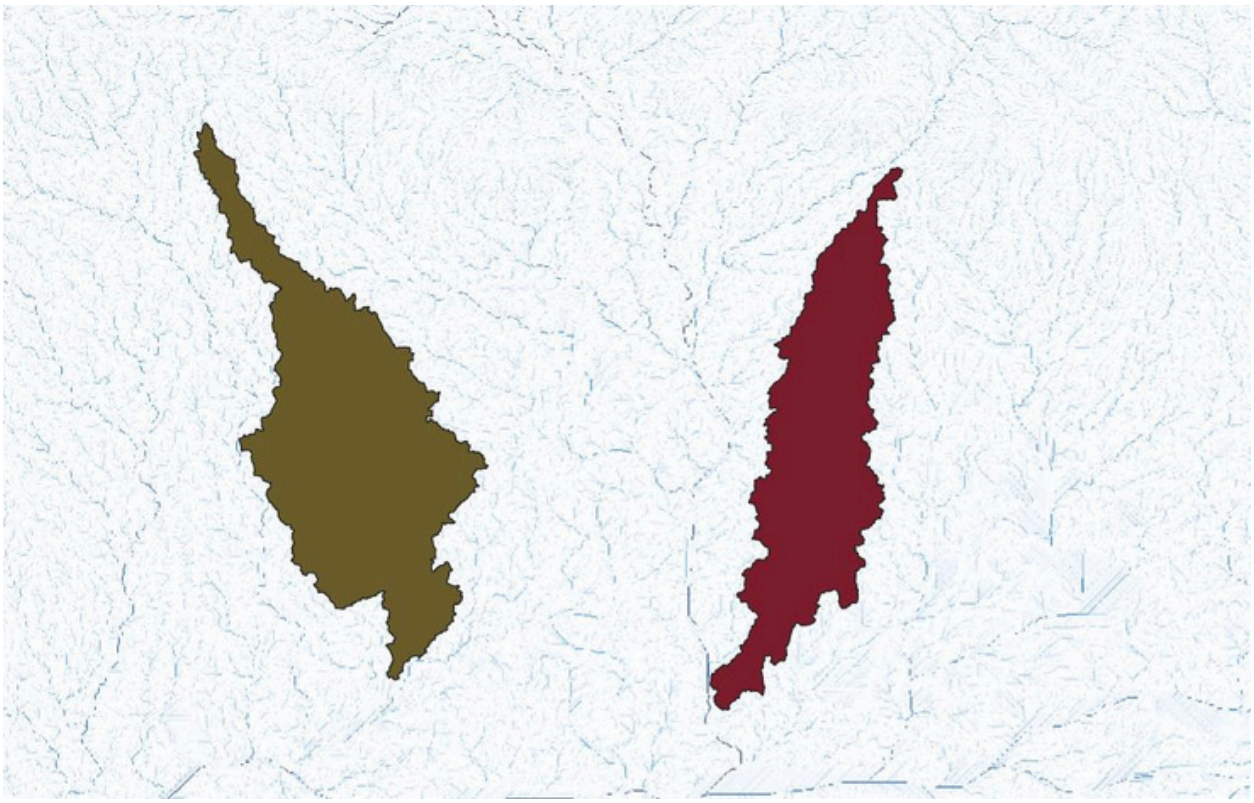
Output:

- Identified villages and towns (also in total numbers) that fall within the delineated watershed.
- Calculated the total area (in sq. km.) of human settlements located in flood-prone zones.
- Compared the identified flood-prone areas with historical flood records to determine high risk regions.

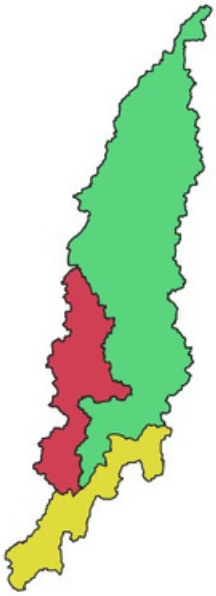
ANALYSIS AND DISCUSSION

Findings:

WATERSHEDS



WATERSHED 1 -SUBWATER SHEDS



AREA

Main water shed : 128.851 sq km

CATCHMENTMERGED@1 — Features Total: 3, Filtered: 3, Selected: 0

	fid	S3	layer	path	Area(km)
1	2	100	S1	C:/Users/hp/De...	128.851
2	1	100	S2	C:/Users/hp/De...	23.665
3	2	100	S3	C:/Users/hp/De...	86.386

WATERSHED -2 SUBWATERSHED

AREA:

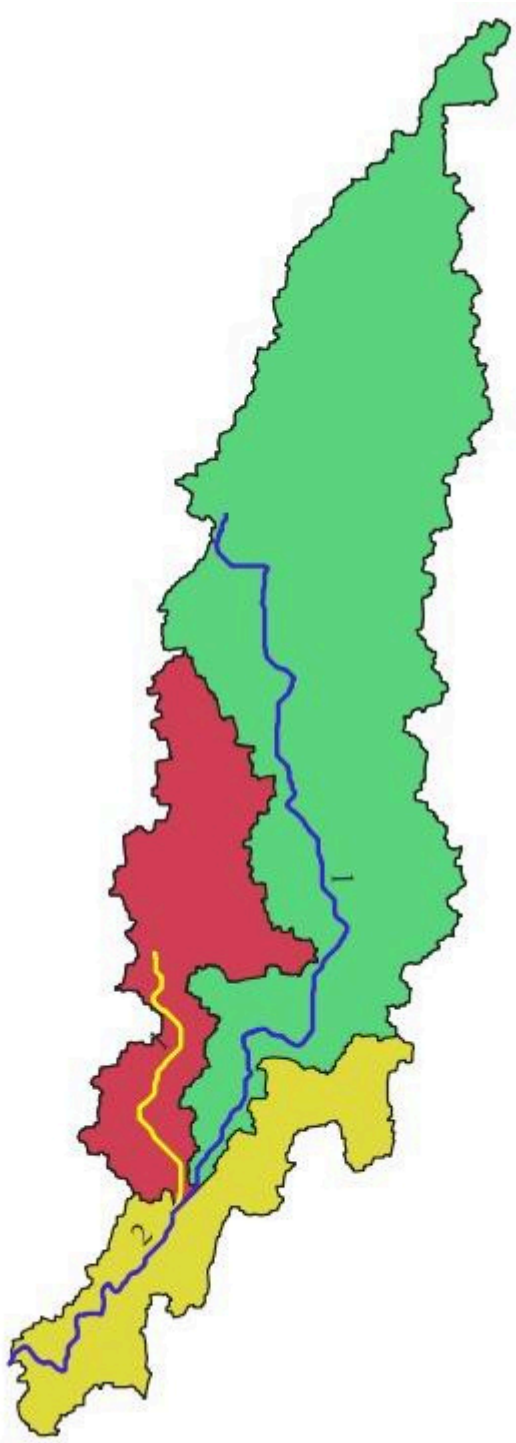
Main water shed: 176.323 sq km

CATCHMENTMERGED@2 — Features Total: 3, Filtered: 3, Selected: 0

	fid	S1	layer	path	Area
1	1	100	S21	C:/Users/hp/De...	176.323
2	1	100	S22	C:/Users/hp/De...	99.414
3	5	100	S23	C:/Users/hp/De...	68.530

STREAM NETWORK ANALYSIS

WATERSHED 1



LENGTH WATERSHED 1

CATCHMENTFULL@1 — Features Total: 3, Filtered: 3, Selected: 1

	SEGMENT_ID	NODE_A	NODE_B	BASIN	ORDER	ORDER_CELL	LENGTH
1	370	391	324	13	1	8	20026.7763090...
2	305	324	299	13	2	9	7584.87492060...
3	333	352	324	13	1	8	6744.95775390...

TOTAL LENGTH

Analyzed field: LENGTH

Count: 3

Unique values: 3

NULL (missing) values: 0

Minimum value: 6744.9577539

Maximum value: 20026.776309

Range: 13281.8185551

Sum: 34356.6089835

Mean value: 11452.202994500001

Median value: 7584.8749206

Standard deviation: 6072.827230432899

Coefficient of Variation: 0.5302758983009135

Minority (rarest occurring value): 6744.9577539

Majority (most frequently occurring value): 6744.9577539

First quartile: 7164.916337250001

Third quartile: 13805.8256148

Interquartile Range (IQR): 6640.909277549999

WATERSHED-2

LENGTH WATERSHED 2

CATCHMENTFULL@2 — Features Total: 3, Filtered: 3, Selected: 1

	SEGMENT_ID	NODE_A	NODE_B	BASIN	ORDER	ORDER_CELL	LENGTH
1	301	319	311	13	2	9	1356.83741150...
2	326	345	319	13	1	8	13828.8056090...
3	364	384	319	13	1	8	18493.8369200...

TOTAL LENGTH

Analyzed field: LENGTH

Count: 3

Unique values: 3

NULL (missing) values: 0

Minimum value: 1356.8374115

Maximum value: 18493.83692

Range: 17136.9995085

Sum: 33679.4799405

Mean value: 11226.493313500001

Median value: 13828.805609

Standard deviation: 7234.095658802842

Coefficient of Variation: 0.6443771404650239

Minority (rarest occurring value): 1356.8374115

Majority (most frequently occurring value): 1356.8374115

First quartile: 7592.82151025

Third quartile: 16161.3212645

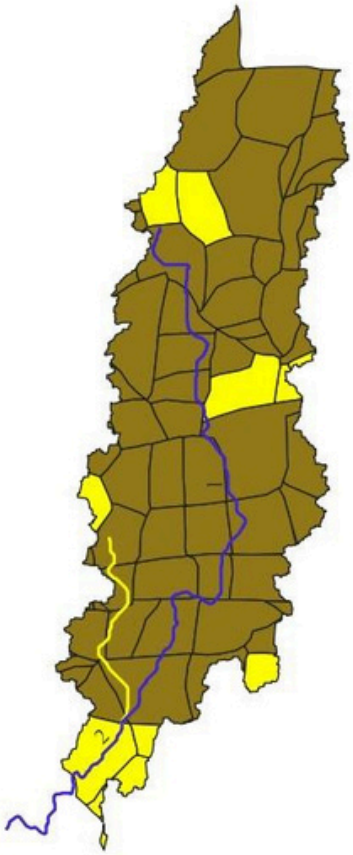
Interquartile Range (IQR): 8568.49975425

FLOOD VULNERABILITY

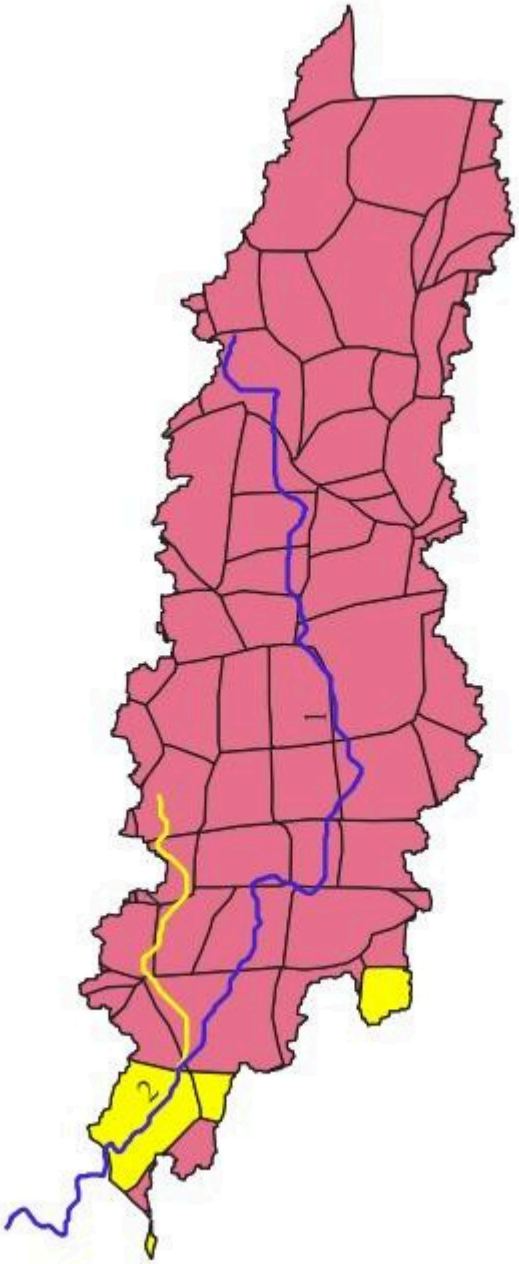
WATERSHED 1

Total town and villages: 71
Vulnerable: 12

Flood prone areas



High risk zones:



TOTAL AREA FLOODED

Analyzed field: AREA

Count: 12

Unique values: 12

NULL (missing) values: 59

Minimum value: 0.009

Maximum value: 3.457

Range: 3.448

Sum: 14.741000000000001

Mean value: 1.2284166666666667

Median value: 0.968

Standard deviation: 1.100906100925758

Coefficient of Variation: 0.8961992545355875

Minority (rarest occurring value): 0.009

Majority (most frequently occurring value): 0.009

First quartile: 0.34450000000000003

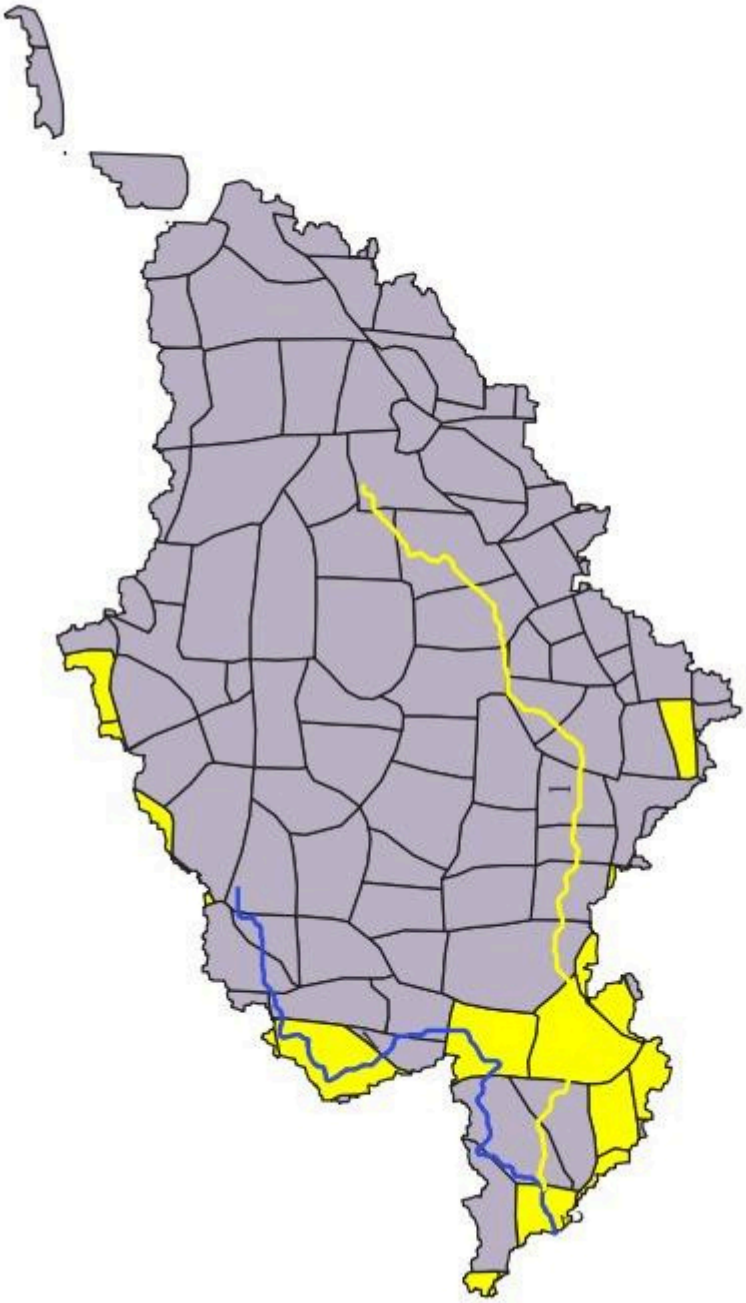
Third quartile: 2.0780000000000003

Interquartile Range (IQR): 1.7335000000000003

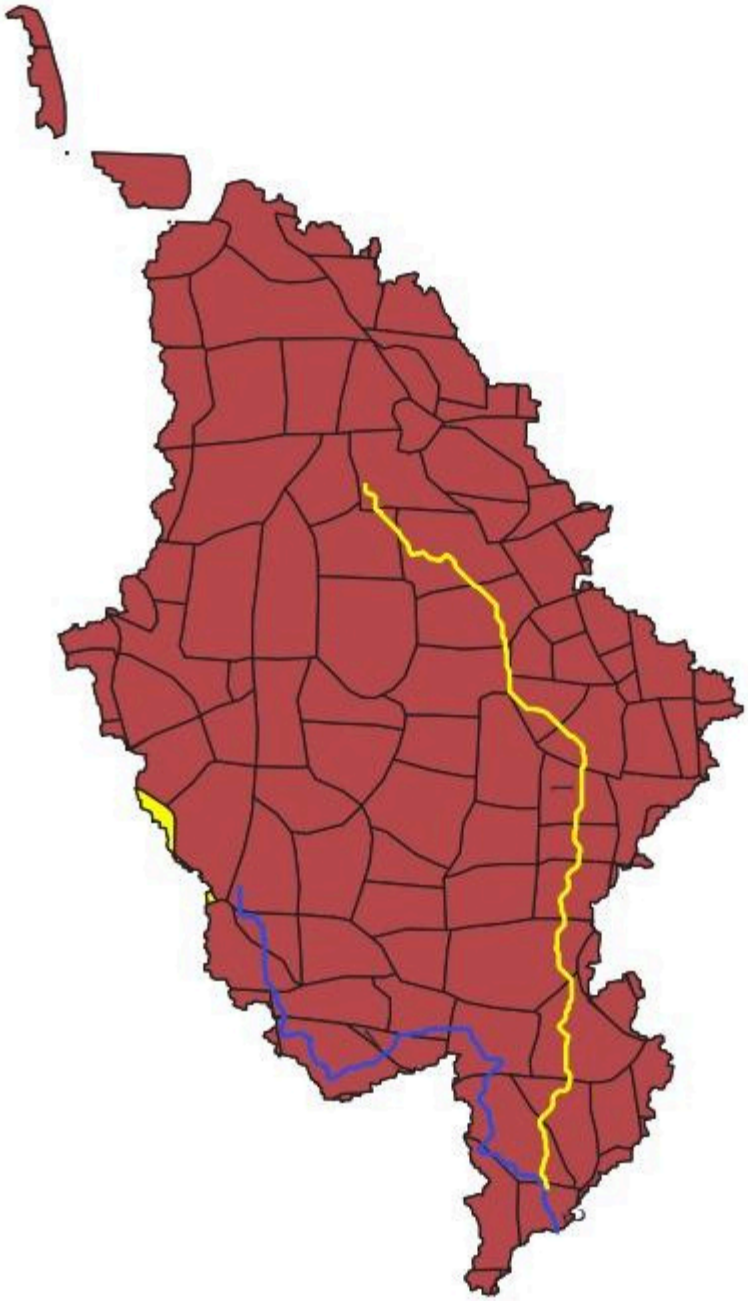
WATERSHED 2

Total town and villages: 111
Vulnerable: 17

Flood prone areas:



High risk zones:



TOTAL AREA FLOODED

Analyzed field: AREA

Count: 17

Unique values: 17

NULL (missing) values: 94

Minimum value: 0.045

Maximum value: 3.048

Range: 3.003

Sum: 15.032

Mean value: 0.884235294117647

Median value: 0.753

Standard deviation: 0.904961262051116

Coefficient of Variation: 1.0234394262153388

Minority (rarest occurring value): 0.045

Majority (most frequently occurring value): 0.045

First quartile: 0.157

Third quartile: 1.1

Interquartile Range (IQR): 0.9430000000000001

ANALYSIS AND DISCUSSION

- To close analysis we could find that major flood prone areas are near the outlets because when water flows from top to bottom it carries forward silts and sand with it. This decreases the carrying capacity of water leading to its spread near the outlets
- Apart from that Assam is an area with hills and plains so during rain water falls through a sharp slope leading to high speed of water which short lengths thus a prone area for flood.
- It has been seen that the majority of the prone areas like Tengabasti, Karaina, Nirai Ati, Kathalguri are also present where the river bends. This might happen due to low infrastructure support near the bending thus, leading to floods.
- Past flood of 2004, 2017 and 2020 indicate recurrent flooding these districts
- This highlights the need for proactive flood mitigation.

CONCLUSION

Sonitpur district of Assam shows very high vulnerability due to its location and vast riverine basins presence in the area. Effective flood mitigation strategies which include both infrastructure and policy making support could help in improve drainage systems which will be essential in reducing flood risks especially in an area where 91% of population's livelihoods relies upon agriculture.

REFRENCES

1. ***Flood & Erosion problems | Water Resources | Government of Assam, India***. (n.d.). https://waterresources.assam.gov.in/portlets/flood-erosion-problems
2. ***India: What's really behind Assam's worsening floods?*** (2024b, July 11). https://www.preventionweb.net/news/india-whats-really-behind-assams-worsening- floods
3. ***A historical understanding of Assam's floods***. (2022, July 29). Economic and Political Weekly. https://www.epw.in/engage/article/historical-understanding-assams-floods
4. ***District Disaster Management Authority Sonitpur | Sonitpur | Government of Assam, India***. (n.d.). https://sonitpur.assam.gov.in/departments/ddma
5. Government of Assam. (2024). ***Flood Contingency Plan 2024-25***. Retrieved from ht t ps://so nit pur . assa m. go v. in/ s it es/de fau lt / file s/pu blic ut ilit y/ Flo o d% 20Co nt ige nc y% 20P lan%202024-25_2.pdf